

CHAPTER TWO

THE EMERGENCE AND DEVELOPMENT OF MANAGEMENT THOUGHT

A. Management in Antiquity and Pioneer Contributors

Management in Antiquity: - Management thought has been shaped over a period of centuries by three major sets of forces. These forces are: Social, economical and political in nature, and they continue to affect management theory even today.

Despite the inexactness and the relative crudity of management theory and science, the development of management thought dates back to the days when people first attempted to accomplish goals by working together in groups. Since pre-historic times people have been managed in groups and organizations. Even the simplest of hunting and gathering bands generally recognized rules and obeyed a leader or a group of decision makers responsible for welfare of the band. As societies grew larger and more complex, the need for organizations and managers became increasingly apparent.

Attempts to develop theories and principles of management, however, are relatively recent. In particular, the industrial revolution of the 19th century gave rise to the need for a systematic approach to management. Some examples in ancient times where management was effectively used include the following:

- **Egypt** - The construction of the Egyptian pyramid (5000-525 BC) is a testimony of the ancient Egyptian organization and managerial abilities. The ancient Egyptians constructed the pyramid by 100,000 labor forces for 20 years on 13 hectares of land using 2,300,000 stones. This construction is equivalent with managing a city with a population of 100,000 for 20 years.

This construction shows how extensively Egyptians used the management functions of planning, organizing, staffing, directing and controlling.

“Comparing the technology and information we have today, managers of those days exceed managers of today.” Sisk

"The best managers in history are the ones who managed the building of the pyramids."

Peter Drucker

- **Romans** - the ancient Romans also provided numerous illustrations of effective management. Perhaps the most famous is the Emperor Diocletian's reorganization of his empire. Assuming his position in A.D. 284, Diocletian soon realized that the empire had acquired an unmanageable form. There were far too many people and matters of importance for the emperor to handle individually. Abandoning the old structure, in which all provincial governors reported directly to him, Diocletian established more levels in the hierarchy. He reorganized the Roman Empire as: Empire into 100 provinces with 13 dioceses and 4 major geographical areas. By doing so he ruled Rome to its best time. The governors were pushed farther down the structure and, with the help of other administrators; the emperor was able to more effectively manage this vast empire. Here the levels of a management are shown clearly: 4 geographical areas ----- 13 dioceses --- 100 provinces
After he divided Rome as such, he appointed 3 people on the divisions and the rest for himself - Delegation of authority.
- **Roman Catholic Church** - was the most successful formal institution in the western civilization. Rome achieved greater colonies using the Catholic Church.

Roman Catholic Church also made important contributions to early management thought. One was the church's wide *use of job descriptions* for its priests, presbyters, and other religious workers. Everyone's duties were clear, and the chain of command (*hierarchy of authority*) that extended from the pope to the laity was created. A second Roman Catholic Church contribution was that of *compulsory staff service*, the requirement that certain members of the church hierarchy seek the advice of the other hierarchs before making particular decisions. A third was the *use of staff independence*, the assignment of certain advisors to key church officials. Since these advisors were not removable by the official they could give advice they considered best, without fear of reprisals from superiors.

In short, the most important contributions of Roman Catholic Church for the development of management are on the areas of:

- ✓ *Hierarchy of authority*: there was a hierarchical structure from Pope - Bishop - priest - laity.
- ✓ *Specialization of activities*: there was a training to be Pope, Bishop, Priest and Laity.
- ✓ *Use of staff managers*:
 - Compulsory staff service
 - Staff independence

- **Greece** - Exhibited a real skill and capacity for management in the operation of trading companies. They recognized the means to maximize output through the use of uniform methods and motion study.
- **Bible** - Exodus 18:13-26; this passage tells how Jethro, Moses', father in-law, observed Moses spending an entire day listening to the complaints and problems of his people. Then Jethro advised Moses that he was doing more than one man should and suggested specific steps to relieve him of his burden. He first recommended that "*ordinances and laws*" should be taught to the people. In modern terms, the origination needed a statement of *policies, rules and procedures*. Second, he commended that leaders be "*selected and assigned*" to be rulers to thousands, and rulers of hundreds, and rulers of fifties and rulers of tens. That was recommending *delegation of authority*. Jethro's third point, that these *rulers should administer all routine matters and should "bring to Moses the important questions,"* forms the basis of a well known control procedure: *the principle of exception*.
- In **Ethiopia**, the construction of obelisks of Axum, Castle of Gondar, Rock Hewn Churches of Lalibela, the Wall of Harrar etc. are good examples that modern management was practiced in ancient time.

Early Management Pioneers (Contributors)

Although examples of management practices go back several thousand years, the development of management as a field of knowledge is much more recent. Much of the impetus for developing management theories and principles grew out of the industrial revolution, which spawned the growth of factories in the early 1800s. With the proliferation of factories came the widespread need to coordinate the efforts of large numbers of people in the continual production of goods.

The challenge posed by the factories brought forth a number of individuals who began to think in terms of innovative ways to run factories more effectively. This group, known as the preclassical contributors to management focused largely on particular techniques that might be applied to solve specific problems. They were followed by individuals who began to develop broader principles and theories that make up major view points, or schools of management: classical, behavioral, quantitative and contemporary. Each of these major viewpoints encompasses several approaches that have contributed to the development of the particular viewpoint.

PRE CLASSICAL CONTRIBUTORS

MAJOR CONTRIBUTORS

A number of individuals in the preclassical period of the middle and late 1800s began to offer ideas that laid the groundwork for broader inquiries into the nature of management that followed. Among the principal preclassical contributors are Robert Owen, Adam Smith, Henry Poor, and Charles Babbage.

1. **Robert Owen** (1771 - 1858) - was a British industrialist and an owner-manager of several successful cotton mills in Scotland.

At that period in history, working and living conditions for employees were very poor. Child workers were common and the standard working day was 13 hrs long. Workers were treated in much the same terms as tools and machines.

Owen was called industrialist and reformer because he was one of the first managers to recognize the importance of human resource in an organization. Because of this he was considered as '*father of modern personnel management*.' His ideas laid the groundwork for human relations movement. Owen was well ahead of his time in recognizing the importance of human resources. He was particularly interested in the working and living conditions of his employees. He said that workers in organizations need special attention and dignity or respect. As a result, he introduced in his organization the following:

- Reduce working hrs from 13 hrs to 10¼ hrs a day,
- Set a minimum hiring age (10 year) to protect children from the abuses of employers,
- Provide meal, housing, and shopping facilities for employees,
- Improved working conditions in the factory

He argued, "Improving the condition of employees would inevitably lead to increased production and profits".

2. **Charles Babbage** (Prof.) Built the first practical mechanical calculator and a prototype of modern computers. Although English mathematician, Charles Babbage (1792-1871) is widely known as "*the father of modern computing*". He also made direct contributions to thinking about management. His management interest stemmed from his difficulties with directing his various projects. He became convinced that the

application of scientific principles to work processes would both increase productivity and lower expenses. Like the 18th century economist Adam Smith, Babbage was particularly enthralled with the idea of work specialization. Work specialization is the degree to which work is divided into various jobs. Smith had concentrated mainly on ways to divide jobs involving physical labor into more specialized tasks, but Babbage carried the specialization idea a step further by recognizing that not only physical work but also mental work could be specialized. Furthermore, he was an early advocate of division of labor principle and the application of mathematics as the efficient use of facilities and materials in production.

Babbage believed that each factory operation should be analyzed so that the various skills involved in the operation could be isolated. Each worker would then be trained in one particular skill and would be responsible only for that part of the total operation (rather than for the whole task). In this way, expensive training time could be reduced, and the constant repetition of each operation would improve the skills of workers and enhance their efficiency.

Division of labor ↘ Specialization/improve skills of workers.
 ↙ Reduce learning time and other expenses.

Emphasizing on efficiency of production, however, Babbage didn't overlook the human element of an organization. He said, "The relationship between management and workers is the reason for the success or failure of the organization." Babbage also had some innovative ideas in the area of reward systems. He devised a profit sharing plan that had two parts, a bonus that was awarded for useful suggestions and a portion of wages that was dependent on factory profits. His ideas foreshadowed some modern day group incentive plans, such as the Scanlon plan in which workers actively participate in offering suggestions to improve productivity and then share in the profits from resulting gains. He understood that a harmonious relation between management and labor could serve to benefit both.

He was an avid proponent of:

- Division of labor
- Economies of scale in manufacturing
- Incentive pay
- Profit sharing
- Application of mathematical concepts in production
- Harmonious relationship between management and workers

Babbage laid the groundwork for much of the work that later became known as *Scientific Management*.

3. Adam Smith: Smith made an important contribution to the development of management thought regarding the impact of division of labor on manufacturing in his book '*The Wealth of Nations*' in 1776. His conclusion was specialization could lead to increased efficiency. This is because:

- Specialization increases the dexterity in every particular work person.
- Specialization saves the time lost in passing from one species of work to another.
- Specialization helps to the invention of great number of machines, which facilitates and bridge and enable one person to do the work of many.

Thus, managers were more interested in the mechanical side of the job. That is, division of labor, coordination of activities, and control of operations.

B. CLASSICAL MANAGEMENT THEORY

The classical management theory emerged during the early years of this century; i.e., it had its foundation during the Industrial Revolution, which began in England with the invention of reliable steam powered machinery and its early applications. This new technology, combined with the concentration of vast amounts of raw material and labor, created a need for management. Its ideas represent the first well-developed framework of management.

The classical viewpoint is a perspective on management that emphasizes finding ways to manage work and organizations more efficiently. It is made up of two/three different approaches: scientific management, administrative management (Classical Organization) theory, and bureaucratic management. This viewpoint is labeled as "classical" because it encompasses early works and related contributions that have formed the main roots of the field of management.

There were two factors that had contributed for the emergence of classical management theory. These are: the research or writings of pioneers such as Charles Babbage, and the evolution of large-scale business management and practices.

The classical approach to management can be better understood by examining it from two perspectives.

These two perspectives are based on the problems each examined. One perspective concentrated on the

problems of lower level managers dealing with the everyday problems of managing the work force. This perspective is known as scientific management. The other perspective concentrated on the problems of top managers dealing with the everyday problems of managing the entire organization. This perspective is known as classical organization theory.

1. SCIENTIFIC MANAGEMENT THEORY

Scientific management is an approach within classical management theory that emphasizes *the scientific study of work methods in order to improve worker efficiency.* It is a systematic study of work which originated in the USA in 1900s i.e., it emphasizes on improving worker efficiency through the scientific study of work. Its objective was to find the most efficient method for performing any task and to train workers in that method.

The emergence of industrial revolution gave rise to factory production system and it in turn caused management to focus on developing the most rational principles for handling its people, machines, materials and money. This challenge took two forms: (1) how to increase productivity (output/input) by making the work easier to perform, and (2) how to motivate workers to take advantage of new methods and techniques. The individuals who developed approaches for meeting these challenges helped lay the foundation for what is known as scientific management.

Scientific management grew up from the research of 5 pioneers. They are Frederick W. Taylor, Frank and Lillian Gilbreth, Henry Gantt and H. Emerson. Since Taylor played the major role, he is called the father of scientific management.

Frederick W. Taylor (1856-1915) is known as "***the father of scientific management.***" Born to a relatively wealthy Philadelphia family, Taylor became an apprentice pattern maker and machinist for a local firm before moving on to Midvale steel. At Midvale, his meteoric rise from laborer to chief engineer in 6 years gave him an opportunity to tackle a serious problem that he had observed- soldiering by workers.

At the beginning of 20th century business was expanding, resources were readily available but labor was in short supply. The primary goal of management during that time was to use the existing labor force efficiently, but there was no skilled manpower. It is at this time that Taylor saw how to use the available resource efficiently. He didn't start new research rather he started to study the problems of the factory production system. The following were some of the problems of the factory system production as he called them:

1. Management had no clear concept of worker - management responsibility.
2. Virtually no effective work standards were applied
3. No incentive was used to improve labor's performance
4. Managerial decisions were made based on intuition, rule of thumb methods or past experience.

He was angry or he was impressed with the degree of "soldiering" - systematic, deliberate delay in performance i.e., deliberately working at less than full capacity and producing less rather than more, due primarily to (1) the workers' fear that they might work themselves out of a job if they produce more, (2) faulty wage systems set up by management encouraged workers to operate at a slow pace. For example, pay by the hour or the day mainly encouraged attendance rather than output. On the other hand, companies that cut incentive pay when workers began to exceed standards also made workers reluctant to excel, and (3) rule of thumb method permitted by managers. These factors led Taylor to conclude that managers, not workers, were responsible for the soldiering because it was up to the management to design jobs and wage systems that could encourage productivity.

To counter the problem of soldiering and solve the problems of factory production system, he timed each element of the work and standardized how much each workman has to produce given the required resources per day or per month. After timing each element of the work he said "the most efficient way of doing the overall work" and came up with the "piece-rate pay system" - the differential rate system - instead of paying similar amount of wage. He began increasing the wage (pay) of each worker who met and exceeded standards or target level of output set for his/her job.

The interest target of Taylor was to utilize human and material resources efficiently and effectively. In other words Taylor was very much obsessed with the idea of maximizing efficiency in an organization in order to maximize profits.

The following are some of the studies he conducted.

1. Time and motion study

Objective: to standardize activities for workers/determine a full-day's work

Time and motion study involves breaking down the task into various elements, or motions, eliminating unnecessary motions, determining the best way to do the job, and timing each motion to determine the amount of production that could be expected per day (without allowances for delays).

In this study he wished to know how long it would and should take a machine or workman to perform a given process to produce a part using specified materials and methods under controlled stations. He used stop - watch system to start and finish the test. This study permitted the determination of practical, relatively precise and reliable standards of output. That is, the study enabled him to set feasible standards per man or machinery usually higher than the average of current performance of that time.

2. Uniform method of routine tasks

Objective: to adjust worker with work.

It is intended to prepare and direct the effort of those responsible for establishing the conditions under which these standards could be set and met. With this objective in mind he worked out such techniques as: instruction cards, order of work cards, material specialization, inventory control systems, material handling standards.

3. Functional foremanship study - which man for which work.

Objective: to scientifically select the best worker for a given job

It was concerned with assuring which man will be best for which job, considering his initial skill and the potential for learning.

4. Individual Incentive

Objective: to determine the appropriate wage or salary

In order to solve the problem of wage systems that encouraged soldiering, Taylor also advocated the use of wage incentive plans. His study reached higher pay would serve as an incentive for workers that would result from the increased productivity. During his (Taylor's) time there was a reduction of rates if the workers earn beyond an acceptable limit. But his view was that, having scientifically measured the worker's jobs and set rates accordingly, then efficient workers should be rewarded for their productivity without limit.

After Taylor conducted the above studies in 1911 he wrote a book called *principles of scientific management*. In his book he outlined four principles, the scope of which represented a combination of mechanical, conceptual, and philosophical ideas:

1. Develop a science for each element of a job, which replaces the old rule-of-thumb¹ method.

¹ Rule of thumb is the usual practice at work organizational level, leaving workers to work according to the initiative of them by managers.

2. Heartily cooperate with the workers so as to insure all of the work being done in accordance with the principles of the science which has been developed.
3. Scientifically select and then train, teach and develop the worker, whereas in the past a worker chose his/her own work and trained themselves as best he could for their own and the company's -prosperity.
4. There is an almost equal division of work and responsibility between management and workers. The management takes over all work for which they are better fitted than workers, while in the past almost all of the work and the greater part of the responsibility were thrown upon the workers.

Taylor testified, however, that in order for these principles to be successful “*a complete mental revolution*” on the part of management and labor was required. Rather than quarrel over whatever profits they were, they should both try to increase production. By so doing, profits would be increased to such an extent that labor and management would no longer have to compete for them. In short, Taylor believed that management and labor had a common interest in increasing productivity.

Frank and Lillian Gilbreth (The first dual career couple) 1868-1972

The Gilbreths were contemporaries of Taylor and part of the original scientific management pioneers. Their accomplishments still stand out for their devotion to a single goal: the elimination of waste and the discovery of the ‘one best way’ of doing work.

Frank started as a bricklayer and then achieved a considerable success as an independent contractor and later as a management consultant. He was determined to learn to lay bricks the ‘right’ way. He began developing his own ideas and work methods. In the end, he was able to develop a method which cut the motions required to lay bricks. In so doing, he identified 17 on-the-job motions and called them ‘THERBLIGS’. (Therbligs is Gilbreth spelt backwards with the transposition of one letter). The on-the-job motions are: search, find, select, grasp, position, assemble, use, disassemble, inspect, load/transport, pre-position, release load, transport empty, wait when avoidable, avoidable delay, rest for overcoming fatigue, and plan and hold. For his work, Frank was named as ‘*The Father of Motion Study*’.

In undertaking his work, Frank was greatly aided and supported by his wife, Lillian. After he died, Lillian was determined to continue his work. She pioneered the field of personnel administration. She argued that the purpose of scientific management is to help people reach their maximum potential by developing their skills and abilities. For her contribution to the field of management, Lillian was known as ***‘The First Lady of Management’***.

Henry L. Gantt (1861-1919)

Gantt worked with Taylor at the Midvale Steel Company and was early proponent of scientific management. If it wasn't for him, scientific management might have been lost in a storm of criticism. Gantt took every opportunity to humanize scientific management such as, inventing a task and bonus system that offered foremen a bonus for every worker who succeeded, which motivates foreman to show interest in their employees and help them achieve as much as they could.

Gantt is perhaps known for his development of ***‘Gantt Chart’*** - a simple graph method of scheduling work according to the amount of time required instead of the quantity of work to be performed.

Contributions of Scientific Management

1. Specialization of activities increases productivity.
2. The methods of scientific management can be applied to a variety of organizational activity, besides those of industrial organizations.
3. The efficiency techniques of scientific management, such as time and motion studies, have made us aware that the tools and physical movements involved in a task can be made more efficient.
4. The stress it placed on scientific selection and development of workers has made us recognize the importance of both ability and training in increasing worker effectiveness.

Limitations of Scientific Management

1. Taylor misread the human element
He equated people (worker) with machine; he said workers are extensions of machines, cogs of machines.
2. Taylor saw only money as a motivator.
He failed to recognize the complex nature of human behavior.

The notions of human behaviors that were prevalent in Taylor's time hampered proponents of scientific management. The then popular model of human behavior was that people were “*rational*” and thus motivated primarily by a desire for material gain. This rationality meant that they would therefore act in a manner best suited to satisfy their economic and physical needs. Thus, Taylor and his followers overlooked the social needs of workers as members of a group and never considered the tensions created when these needs were frustrated. They assumed one had only to tell people exactly what to do to increase their earnings and they would go right ahead and do it, as “rational” people should. Financial gain, while significant, is not the only thing that matters to workers.

3. The human desire for job satisfaction was overlooked. Workers usually are more willing to go out for strike over job conditions rather than salary and to leave a job if they were unhappy with it.

4. The application of scientific principles was not smooth.

Give money to workers and then work. If they don't work, penalize if not fire them out.

5. Taylor didn't consider the organization as a whole but the production or the technical level.

6. Too often, increase in productivity lead to lay-offs or changes in piece rates which leave workers producing more output for the same income. The higher wages and better working conditions enjoyed by today's workers don't result solely from the voluntary redistribution of increased profits by management instead because of the tremendous growth of unionism.

2. Classical Organization Theory

Scientific management was concerned with increasing the productivity of the shop and the individual worker. The other branch of classical management- classical organization theory-grew out of the need to find guidelines for managing complex organizations. That is, the classical organization theory focused on the management of the entire organization unlike the scientific management theory, which focused on production. Classical organization theory had two major purposes: (a) develop basic principles that could guide the design, creation and maintenance of large corporations, and (b) identify the basic functions of managing organizations. Henri Fayol, Lyndell Urwick, Chester Bernard, Max Weber and others contributed towards the development of classical organization theory. Henri Fayol's work will be discussed here because his ideas clearly explain classical organization theory.

Henri Fayol (1841-1925)

A French industrialist and a well-known contributor to the administrative management theory, was born to a middle class family near Lyon-France. He is acknowledged as the prominent contributor and founder of the classical organization theory.

On the basis of his experience as a top-level manager, Fayol was convinced that it should be possible to develop theories about management that could then be taught to individuals with administrative responsibilities. To him management (administration) was not a personal talent but a skill that could be taught. His efforts toward developing such theories were published in *General and Industrial Management*, which originally appeared in monograph form in 1916 but attained prominence in the United States after a second translation appeared in 1940.

Henri Fayol is the one who identified:

A. The major types of **activities involved in an industry or a business as:**

- *Technical* - producing and marketing products
- *Commercial* - buying raw materials and selling products
- *Security*: protecting employees and property
- *Financial* – search for and optimum use of capital
- *Accounting* - recording and taking stock of costs, profits, and liabilities, keeping balance sheets, profit and loss statements, etc
- *Managerial* – planning, organizing, commanding, coordinating and controlling

These six elements, he said, will be found regardless of whether the undertaking is simple or complex, big or small.

Fayol's primary focus was the managerial activity, because he put managerial skill had been the *most neglected, least understood* but the *most crucial* aspect of business operations. He defined managing in terms of the five functions: planning, organizing, commanding, coordinating and controlling which all are similar to the present-day five managerial functions of Planning, Organizing, Staffing, Directing, and Controlling.

B. Management as a separate field of study

Fayol said management is a discipline worth studying because he believed that managerial ability could be applied to the home, the church, the military, the politics, and industry. Adding he said there is a need for the introduction of formal managerial training schools.

C. General management principles

In his managerial experience Fayol attempted to systematize the practice of management to provide guidelines and directions to other managers. Part of his thinking was expressed in the 14 principles of effective management. Noting that the administrative function was concerned only with the human part of an undertaking, Fayol hastened to explain in his monograph that he employed the word **principles, not laws or rules**, because of the flexibility required in applying such concepts to people. In his words, he said,

“I prefer the word principles in order to avoid any idea of rigidity, as there is nothing rigid or absolute in administrative matters; everything is a question of degree. The same principle is hardly ever applied twice in exactly the same way, because we have to allow for different and changing circumstances, for human beings who are equally different and changeable, and for many other variable elements. The principles, too, are flexible, and can be adopted to meet every need; it is just a question of knowing how to use them.”

The 14 principles that Fayol felt he had occasion to use most frequently were:

1. Division of labor

Division of work (labor) encompasses three basic concepts:

1. Breaking down a task into its components.
2. Training workers to become specialist in specific duties, and
3. Putting activities in sequence so one person's efforts build on another's

The theory is that the fewer the tasks a person does in his job, the more efficient, skilled and effective he becomes. Yet there are limitations to how much that work should be divided.

2. Authority and responsibility

Authority is the right to give orders, to exact obedience. Responsibility, on the other hand, is a sense of obligation that goes with authority or a reward or penalty accompanying the use of this power. Authority should be delegated only to subordinates who are willing to assume commensurate responsibility. In later years this principle would be called parity of authority and responsibility, indicating that the two should always be equal.

3. Discipline

Members in an organization need to respect the rules and agreements that govern the organization. To Fayol, discipline will result from good leadership at all levels of the organization, fair agreements (such as provision for rewarding superior performance), and judiciously enforced penalties for infractions. Fayol saw the necessity for discipline and precise and exact obedience at all levels for the smooth running of a business.

4. Unity of Command

An employee should receive directives from only one superior. One person should have one boss. S/he should receive orders from one boss and resort to the same boss.

Fayol believed that when an employee reported to more than one superior, conflicts in instructions and confusion of authority would result, i.e., violating this principle undermines authority and jeopardizes discipline and stability.

5. Unity of Direction

Where there are a group of activities with the same basic objective there must be one coordinated plan to accomplish the coordination of effort, and one person at the head of such coordination: **ONE HEAD, ONE PLAN, ONE SET OF OBJECTIVES**. This improves coordination and ensures that energies are channeled in the proper direction. Unity of command is related to personnel whereas unity of direction relates to the organization of the 'body corporate' and asserts that all operations that have the same objective move in the same direction under one unified plan and under the command of one superior.

6. Subordination of individual interest to general interest/common good

In any undertaking, the goals and interests of an organization must take precedence over those of individuals or groups of employees. The overall interest of the firm is more important than the interest of any person or group of people who work for it.

7. Centralization

Centralization and its counter part, decentralization, mean how much authority is concentrated at the top of the organization or dispersed throughout the management hierarchy. Fayol believed that managers should retain final responsibility but also need to give their subordinates enough authority to do their jobs properly.

The problem is to find the optimal amount of centralization. The appropriate degree of decentralization or

centralization depends on the situation and includes such factors as the nature of the task and the abilities of subordinates.

8. *Remuneration of personnel*

Compensation or wages for the work done should be fair and equitable to both to workers and the organization. Payment plans shouldn't lead to over payment but the amount and method of payment should be fair to reflect the cost of living, general economic condition, the demand for labor, the economic state of business and the value of employees.

9. *Scalar Chain (Chain of Command)*

Scalar chain is the line of authority that extends from the top to the bottom of an organization and defines the communication path. All organizational requests and directives must follow this chain. This must be the route to be followed by all communications via every link in the chain that starts from the bottom level of the organization to the ultimate authority. This preserves the integrity of the hierarchy, and ensures the unity of command.

The only time departure from the chain of command can be tolerated is when the welfare of the organization is at stake. However, Fayol recognized the problem of red tape in a large organization and the resulting inadvisability of always taking the long formal route. To overcome this problem, Fayol prescribed the gangplank principle. People at the same level of the hierarchy should be allowed to communicate directly, provided that they have the permission from their superiors to do so and that they tell their respective chiefs afterward what they have agreed to do.

10. *Order*

Order is best defined as “a place for everything (everyone) and everything (everyone) in its/his place. Materials and people should be in the right place at the right time. People in particular should be in the job or positions most suited for them.

11. *Equity*

In dealing with subordinates, managers should be friendly, fair, kind and lawful. Kindness and justice on the part of managers will help employees to be loyal and devoted workers.

12. *Stability of tenure of personnel*

Trained and experienced workforce and management are crucial for organizational success. Both managers and workers need time to learn and master their jobs. The longer these people stay in the organization or on

their jobs, the higher their belongingness to the organization or/and the greater their effectiveness will be. If they leave or are removed within a short time, the organization will lose the belongingness that it could accrue and the learning time incurred. Therefore, labor turnover should be minimized and stability nurtured. Note, however that some turnover is expected and is desirable. Factors like retirement, death, dismissal or demotion because of incompetence could lead to turnover.

13. Initiative

Members of an organization should be given the opportunity to demonstrate their creativity, exercise their judgment, chart out their own plans of discharging their responsibility within the bound of due respect for authority and discipline. To Fayol, the keenest interest for an intelligent executive is to be able to scarify his personal vanity and instill those under him with the attribute to bake initiative. This will provide organizational members with a platform for realizing their capabilities to the fullest and appreciate the active part they are playing. The organization in turn will enjoy high quality and forward-looking decisions and better performance.

14. Esprit de corps

The gist of this principle is that 'in union there is strength.' Hence, workers, management as well as employers should work as a team. It is the responsibility of management to promote the spirit of cooperation rather than the spirit of divide and conquer. Fayol made the point that it would be wise to divide the enemy forces but to apply the same strategy on ones own team is a folly and calamitous. Fayol also said that oral communications should be preferred to written directives and explanations whenever possible for written communications could lack speed or clarity and breed dubiety and reservation rather than trust and cooperation.

Contributions of Classical Organization Theory

1. The position Fayol took in distinguishing management as a discipline was worth studying.
2. The fourteen basic management principles he developed.
3. The five elements of administration, which with minor modifications today, are called functions of management.

Limitations of Classical Organization Theory

1. Some of the principles are rigid- e.g. chain of command, unity of command.
2. The 14 principles are applicable in a relatively stable and predictable environment and hence they are less appropriate in today's turbulent environment.

3. The principles of classical organization theorists are too general for today's complex organizations. It doesn't provide guidance for deciding which principle should take precedence over the other. E.g. unity of command principle seems to contradict the principle of division of labor (specialization), which states efficiency will be increased if one task is divided among the members of a group.

Contribution and Limitation of Classical Management Theory

Contributions

1. Classical management theory had laid the foundation for later developments in management theory. It laid the groundwork for modern management theory.
2. It identified key management processes, functions, principles and skills that are recognized as such today.
3. It focused attention on management as a valid subject of scientific enquiry, i.e., as a separate field of study.
4. It was the first attempt to study management from scientific perspective.

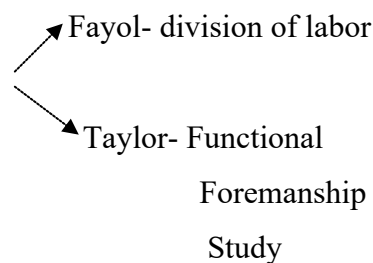
Limitations

1. The theory is more appropriate for stable, simple organizations than for the today's dynamic and complex organizations.
2. It often prescribes universal principles that are not really appropriate in some settings. E.g. chain of command
3. Many writers viewed organizations from mechanistic point of view than from social point of view.
4. Many writers viewed employees as tools than human beings. E.g. Taylor
5. Economic motivation was seen as the only means to motivate workers to produce more.

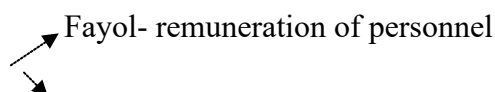
Similarities and differences between scientific management theory and classical organization theory

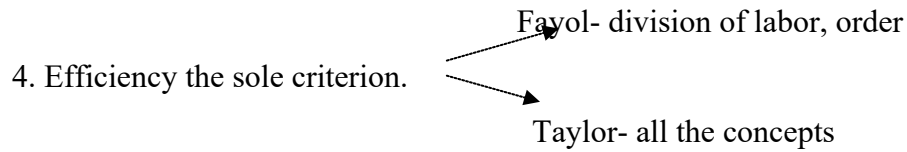
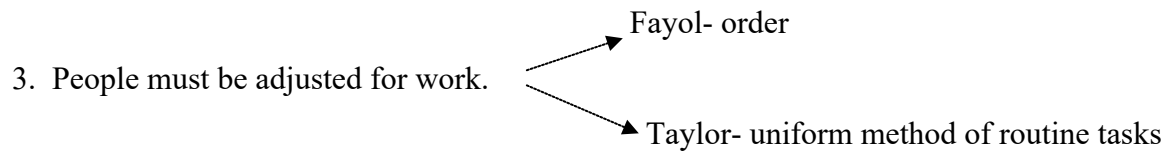
Similarities

1. Both of them focused on division of labor.



2. Economic (job) security motive only.





Differences

Classical organization theory focused on the management of the entire organization and on the problem of the top level management whereas scientific management theory focused on production level/shop level/ technical level and on the problem of the lower-level management.

Frederick W. Taylor (1856-1915) Classic Scientific	Dedicated to the refinement of scientific principles and their applications to production efforts. Taylor was an early pioneer in time and motion study, whose objective was to remove fatigue and improve efficiency and output. He advocated a system of pay based on output by an individual worker (piece-rate pay). Taylor is often called the Father of Scientific Management.
Henry L. Gantt (1861-1919) Classic Scientific	Promoted concern for workers and a movement away from authoritarian management. He introduced a task and bonus system of paying wages that paid a fixed for below standard work, a bonus if standards were met, and an additional amount when standards were exceed. Gantt asserted that methods improvements are best when they are willingly adopted by the people who must apply them. He was the inventor of the Gantt chart for programming production.
Henri Fayol (1841-1925) Classical Administrative	Applied the scientific management concepts to top management and administration. Fayol developed the fourteen basic principles of management that underline all managerial tasks. He identified five basic management functions: planning, organizing, commanding (directing), coordinating, and controlling.

3. Behavioral Management Theory

The behavioral management theory emerged partly because practicing managers found the classical management approach/theory didn't achieve complete production and work place harmony; because classical management theory viewed organizations from the mechanistic view point and considered workers as cogs.

Classical management theorists focused on controlling and standardizing the individual behavior, but it is difficult to standardize group or individual human behavior. Practicing managers got problems in managing organizations because subordinates and workers weren't behaving as expected by scientific/classical management theory. As a result behavioral management theory merged opposing the ideas of classical

management theory, and emphasized on human relations- a general term used to describe the ways in which managers interact with their subordinates- based on social environment of work, individual and group behavior and interpersonal relationships.

They used concepts from psychology, sociology and anthropology to assist managers understand human behavior in the work place. They focused on motivation, communication, work group formation and leadership.

Behavioral management theory was stimulated by a number of writers and theoretical movements. Among writers Abraham Maslow, Douglas Mc Gregor and Elton Mayo were well-known. Of these Elton Mayo was the most prominent one. The behavioral school of management had its origins in industrial psychology and sociology. It emphasizes the interactions of people in an organization in order to understand the practice of management.

The human relation movement marked the daybreak of behavioral science theory and historically it represented a reaction to the dehumanization aspects of scientific management theory. While the former emphasized human relationships in organization, the later (task management) focused on the management of work (the task) and aimed to achieve total efficiency and work place harmony: but in actuality disorder and animus instead of agreement and friendliness continued. Despite then adoption of scientific management, management as well as labor still encountered serious problems because one seldom behaved as the other wanted to be. In short, scientific management stood far from success in accomplishing its mission of achieving a mental revolution in the minds of both the employers and the employees. Consequently, since the human problem grew as crucial as the machine problems it appeared essential to find a means that could help managers become effective in dealing with people and thereby increasing people's productivity. To this end a series of experiments were conducted in 1920s and early 1930s and these experiments pointed the way to new approach to the problem of productivity.

The Hawthorne studies

The Hawthorne studies (1924-1932) had their roots in the logic of scientific management. The initial purpose of these experiments was to study the effect of physical factors such as illumination, rest periods, length of working days, and the payment schemes up on productivity; because classical management theorists believed that physical factors are determinants of workers productivity. The studies were conducted at the Hawthorne plant of Western Electric Company in Illinois, USA. The Hawthorne studies consist of four major experiments.

1. Illumination Experiments

The intention of this experiment was to learn if there was any correlation between intensity of light and productivity. To this effect two groups of women were taken: the experiment group - one subjected to variety in the intensity of light and the other a controlled group which was exposed to constant illumination intensity. But in the end the researchers were bewildered by their experiment since productivity not only constantly increased in both the experimental and controlled groups following the increase in the intensity of illumination in only the experimental room but also kept on ascending in both rooms though the light was diminished so that it was barely enough to see. After seeing this puzzle researchers concluded that illumination has little or no effect on productivity.

2. Relay Assembly Test Room Experiment

In the relay assembly test room experiment, Mayo and his associates placed two groups of six women (five assemblers and a layout operator) with an observer who was to record everything that happened and to maintain a friendly atmosphere in separate rooms. The six workers were told that the experiment was not designed to boost production but merely to study various types of working conditions so that the most suitable environment could be ascertained. They were instructed to keep working at the regular pace. Researchers allowed the groups to choose their own rest periods (were allowed to leave their work station without permission) and to have a say in other suggested changes.

A number of variables were tried: salaries, coffee breaks (rest periods), refreshments, workday and workweek, temperature, and noise. In one room job conditions were varied and in the other they were not. Output went up in both the test room and controlled room regardless of how the factors under consideration were manipulated.

However, when these changes were later terminated and original conditions reestablished, output still remained high, indicating that the change in conditions was not the only reason for the increase in output. Some investigators hypothesized that the increases were related not to the rest pauses or shorter working hours but to the improved outlook that the workers had toward their work.

Mayo and his associates concluded that such physical changes have no significant effect on productivity, and said the change in supervisory management was the major reason for the increase in productivity in the relay assembly test room study. That is, assemblers were exercising relative self-direction and control, were subjected to the observation of a single man, were allowed to talk freely among themselves and as a result close relationship was maintained and a new social environment was created. In order to gather

information on this idea, the management decided to investigate employee attitudes and the factors to which they could be traced. The result was the massive interviewing program.

3. The Massive Interviewing Program

After the first two phases, the researchers concluded that their attempt to relate physical conditions of the job to productivity did not produce any significant results. So they postulated that the human element in the work environment apparently had a significantly greater impact on productivity than the technical and physical aspects of the job. On the basis of their extensive interview program, the researchers proposed that the work group as a whole determined the production output of individual group members by enforcing an informal norm of what a fair day's work should be.

Over 20,000 interviews were conducted in the third phase of the studies. The interviews begun by asking employees direct questions about supervision and the work environment in general. Although the interviews made it clear that answers would be kept in strict confidence, the responses to questions were often guarded and stereotyped. The approach was therefore changed from direct to indirect questioning. The employees were free to choose their own topics. A wealth of information about employee attitudes resulted. The researchers realized that an individual's work performance, position and status in the organization were determined not by that person alone but also by the group members. Peers had an effect on individual performance. In order to study this more systematically, the research entered its fourth and final phase: The Bank Wiring Observation Room Study.

4. The Bank Wiring Observation Room Study

To test the premise formulated at the conclusion of the interview program, the researchers conducted a final experiment. The procedure in this part of the study was similar to that used in the relay assembly test room, except that nine males who assembled terminal banks for telephone exchanges were selected.

This experiment focused on the effect of a group piecework incentive pay. The assumption was that the workers would seek their own economic interests by maximizing their productivity and that faster workers would pressure the slower ones to improve their efficiency. However, the researchers found that pressure was actually a form of social behavior. In order to be accepted in the work group, the worker had to act in accordance with group norms and be a "rate buster" by overproducing or a "chiseler" by under producing. The group defined what constituted a day's work, and as soon as they knew that they could reach this output level, they slacked off. This process was more marked among the faster workers than the slower ones.

The researchers concluded that the work group set the fair rates for each of its members. They found no relationship between productivity and intelligence, dexterity, and other skills. They concluded that the wage incentive plan was less important in determining an individual worker's output than was group acceptance and security.

Findings and implications of Hawthorne

The Hawthorne studies constituted the single most important foundation for the behavioral approach to management. The conclusions drawn from them were many and varied.

1. Physical working conditions did not seem to explain the changes that were related in productivity.
2. There are other factors other than physical factors and monetary incentives, which affect productivity. These factors are social and psychological in nature.
 - Social environment:
 - Ability to talk to each other.
 - The right to choose their rest periods.
 - The right to leave the workstation without permission.
 - The right to have a say in suggested changes.
 - Psychological conditions
 - Since they were selected as a member of the study group they felt social acceptance, recognition, and social importance.

From this we can understand that human beings are social beings rather than rational, economic beings. In addition we can understand that to be successful managers should understand people along with the reward systems, machines and tools (socio-technical aspect). That is, Human aspect must match with social aspect.

3. Workers are not motivated by the bodily needs only but also by social and psychological needs.
4. A kind of managerial leadership capable of understanding individual and group behavior and that would serve them through such skills as motivation and communication is necessary.

5. **Hawthorne Effect:** a second finding, and probably the most widely cited, is the Hawthorne effect, which is simply the observation that when people know that they are being watched, they will act differently than when they are not aware of being observed. The Hawthorne effect refers to the possibility that individuals singled out for a study may improve their performance simply because of the added attention they receive from researchers, rather than because of any specific factor being tested in the study. Applying this concept to the increase in productivity in the relay room, many modern psychologists contend that it was not the changes in the rest pauses that led to increased output but the fact that the workers liked the new situation, in which they were considered to be of some significance. The attention given them led them to increase their output. The Hawthorne effect thus seemed to lead to the decline in revery, but further investigation indicated that it was apparently not the only factor involved.

Contributions and Limitations of Behavioral Management Theory

Contributions

1. It has changed managerial thinking. Managers are now more likely to recognize the importance of people and to view workers as valuable resources than mere tools. Mayo had rediscovered Owen's century old dictum.
2. It has provided important insights into motivation, group dynamics and other interpersonal processes in organizations.

Limitations

1. The complexity of individual behavior makes prediction of that behavior difficult
2. Contemporary research findings by behavioral scientists are not often communicated to practicing managers in an understandable form.

Modern approaches to management

Traditional organizational theories used a highly structured closed system approach. But modern theories have moved towards the open system approach. That is, the classical and neo classical approaches focused on the internal management of the organization. They didn't take into consideration the effect of the external environment on the organization and vice versa.

While the classical and behavioral approaches continue to make contributions to management, other viewpoints also have emerged. These are contemporary in the sense that they represent recent major

CHAPTER 2: MANAGEMENT THOUGHTS

innovations in thinking about management. Two of the most important contemporary viewpoints are the systems and contingency theories.

1. Systems Theory

The systems theory approach is based on the notion that organizations can be visualized as systems. A System is a set of interrelated parts that operate as a whole in pursuit of common goals. The systems approach to management views organizations and the environment within which they operate as sets of interrelated parts to be managed as a whole in order to achieve a common goal.

According to the systems approach, an organizational system has four major components.

- a. **Inputs** - are the various human, financial, equipment and informational resources required to produce goods and services.
- b. **Transformation process** - are the organization's managerial and technological abilities that are applied to convert inputs in to outputs.
- c. **Outputs** - are the products, services and other outcomes produced by the organization.
- d. **Feedback** - is information about results and organizational status relative to the environment.
It is a key to system control.

Contemporary systems theory finds it helpful to analyze the effectiveness of organizations according to the degree to which they are open or closed. There are two types of systems: open and closed

Open system

It is one that continually interacts with its environment and therefore is well informed about changes within its surroundings and its position relative to these changes. The open system engages in such interactions in order to take in new inputs and learn about how its outputs are received by various important outside elements.

It is a system which interacts with external environment and is dynamic and adaptive with the change in the environment. It has permeable boundary between itself and the broader supra system. E.g. social systems, biological systems

Organizations that operate closer to the open end of the system share certain characteristics that help them survive and prosper. Some are: negative entropy, differentiation, and synergy.

Entropy - refers to the tendency of systems to decay over time. Or, it states that systems will decay overtime if they don't interact with the environment. It is a natural process by which all things tend to breakdown or die. If a system does not bring in or receive inputs and energy from its environment, it will eventually cease to exist.

In contrast, **negative entropy** is the ability of open systems to bring in new energy in the form of inputs and feedback from the environment in order to delay or arrest entropy, the decaying process. Organizations must monitor their environment, adjust to changes, continuously bring in new inputs in order to survive and prosper.

Differentiation - is the tendency of open systems to become more complex. The increased complexity usually stems from the addition of specialized units to handle particularly troublesome or challenging parts of an environment.

Synergy - the main gist of this concept is "the whole is greater than the sum of its parts." This means that an organization ought to be able to achieve its goals more effectively and efficiently than would be possible if parts are operated separately. It emphasizes on the importance of working together in a cooperative and coordinated fashion. The simultaneous action of different parts of an open system functioning in a harmonious and integrated manner produces more total effect than the sum of the separated effort of individual parts.

Steady state - the balance to be maintained between inputs flowing in from the external environment and the corresponding outputs returning to it. Organizations should adapt to environmental changes. Steady state is the tendency of maintaining equilibrium condition by making constant and proportional adjustment in response to changes in its environment. An organization in steady state is not static, but in dynamic form of equilibrium.

Closed system

A closed system is a system that does little or no interaction with its environment and receives little feedback. It has rigid boundary.

E.g. physical systems, mechanical systems.

Subsystems - the parts that make up the whole of the system. Each system may be a subsystem of a still larger whole until we reach the larger supra system.

Department, plant, industry, national economy, the world system sequential relationship can show us the system-subsystem formation.

- According to the systems viewpoint, managers are likely to be more successful if they attempt to operate their units and organizations as open systems that are carefully attuned to the factors in the environment that could significantly affect them. Hence, the system approach views organizations as open systems having interdependence and interactions between the organization and its environment and among various subsystems to exchange information and energy.

CONTINGENCY THEORY

The classical theorists like Taylor and Fayol, were attempting to identify " the one best way" for managers to operate in a variety of situations. **If universal principles could be found, then becoming a good manager would essentially involve learning the principles and how to apply them.** Unfortunately, things were not simple. Researchers soon found that some classical principles such as Fayol's unity of command could sometimes be violated with positive results. Consequently, contingency theory began to develop. ***Contingency theory is a viewpoint that argues that appropriate managerial action depends on the particular parameters of the situation.*** Hence, rather than seeking universal principles that apply to every situation, contingency theory attempts to identify contingency principles that prescribe actions to take depending on the characteristics of the situation. Contingency theory suggests that appropriate managerial behavior in a given situation depends on or is contingent on a wide variety of elements. Managerial decisions must be specific for specific situations by recognizing the uniqueness of the environment. It states, "Nothing is best for all situations."

The basic idea of contingency theory was that there is no one best way of managing. Every organization is unique, existing in a unique environment, with unique employees and unique goals. Managerial practices and technique that are appropriate in one area might not be appropriate in another. This is because the world is too complex to be managed by a single approach in all situations.

INTEGRATIVE APPROACH

The objective of the integrative approach is to have integration between different management theories. It states that classical, behavioral and other schools of management are supplementary and the systems and contingency approaches can help to integrate the various schools of thought. The initial premise of the integrative approach is that before attempting to apply any concepts or ideas from the various schools of management thought we must recognize the interdependence of units within the organization, the effects of environmental influences, and the need to respond to the unique characteristics of the situation that arises in terms of its unique characteristics.